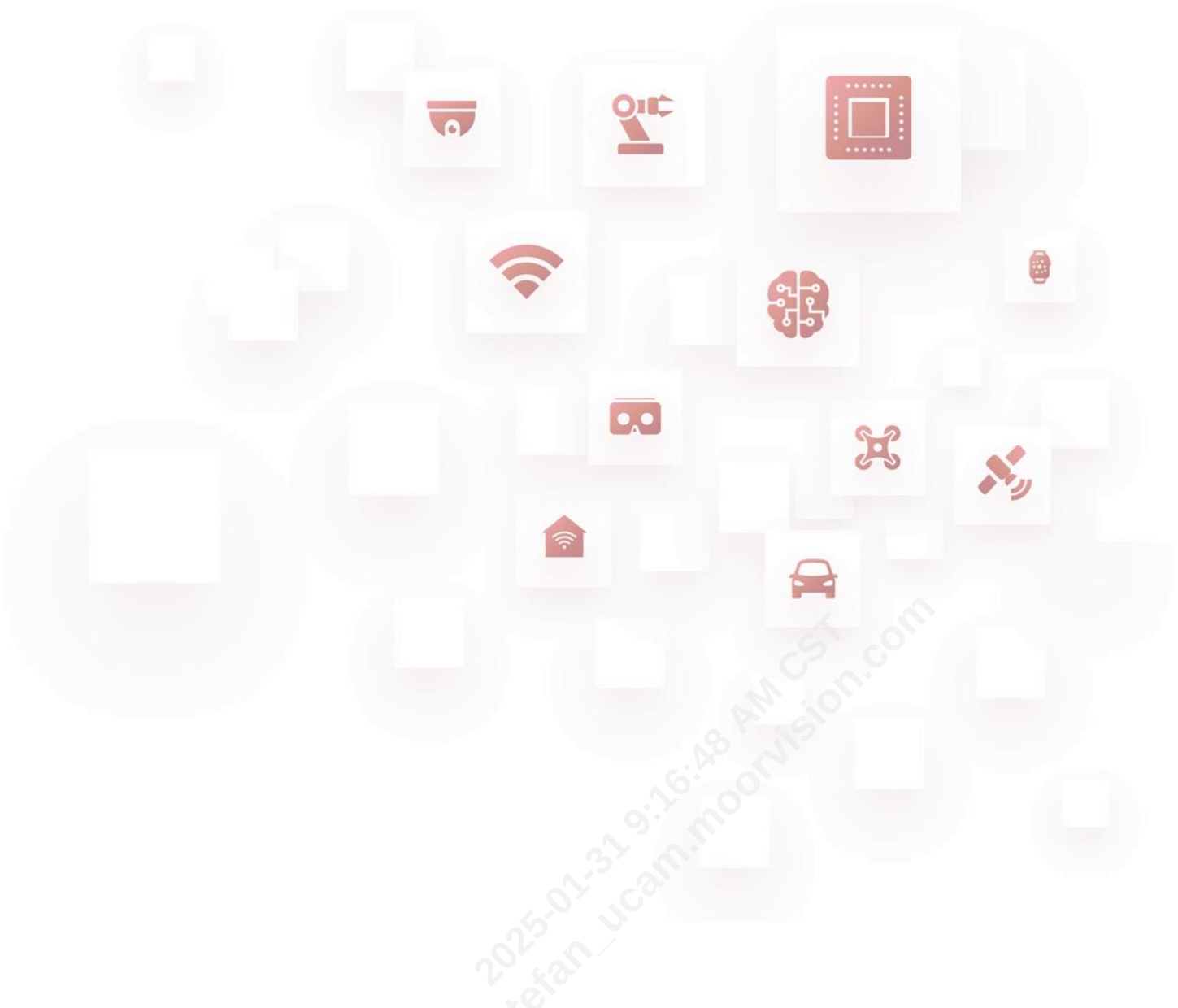
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Thundercomm TurboX C410/C610 Open Kit Linux Android

Software Release Notes

Rev. V1.0
July 17, 2023

DN: tc-gla-2410
SDK Version: LA2.0.Post-CS1.r002002

Revision History

Revision	Date	Author	Description
1.0	July 17, 2023	Ranran Cong	Initial release.

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About This Document

- Illustrations in this documentation might look different from your product.
- Depending on the model, some optional accessories, features, and software programs might not be available on your device.
- Depending on the version of operating systems and programs, some user interface instructions might not be applicable to your device.
- Documentation content is subject to change without notice. Thundercomm makes constant improvements on the documentation of the products, including this guidebook.
- Function declarations, function names, type declarations, attributes, and code samples appear in a different format, for example, `cp armcc armcpp`.
- Code variables appear in angle brackets, for example, `<number>`.
- Button, tool, and key names appear in bold font, for example, click **Save** or press **Enter**.
- Commands to be entered appear in a different font; on the host computer use $ as shell prompt, while on the target device use # as shell prompt, for example,

```
$ adb devices
# logcat
```

- Part of the code that does not contain instructions appear in a different format, for example,

```
SUBSYSTEM=="usb", ATTR{idVendor}=="18d1", MODE="0777", GROUP="adm"
```

- Folders and files are formatted in italic, for example, *turbox_flash_flat.sh*.

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Chapter 1. Introduction

This document provides information on how to obtain, build, and program QCS410/610 LA (Linux Android) SDK (Software Development Kit) into the TurboX C410/C610 Open Kit.

➲ **NOTE:** "TurboX" referred to herein is the English text of our registered trademark TURBOX.

The Linux Android Software Development Kit supports the Android operating system on the Thundercomm TurboX. The goal of the LA SDK is to provide a simple and flexible development platform to rapidly build IoT products with the QTI chipsets. The LA SDK provides the build framework, software components, tools, and sample applications to evaluate the platform and build software images for the target product.

➲ **NOTE:** This release only supports the 4GB TurboX C610/C410 Open Kit. Running it on a 2GB version would cause certain functions to be unavailable due to insufficient memory.

Table 1-1. TurboX C410/C610 platform specifications

SOC	SKUx		Standard SKU[1)]	OS	Thundercomm Product No. (TPN)	Model
QCS610	SKU1	2+16GB	Yes	LA[2)]	C610-D21-369PWXX C610-D21-ACXPW00 (LA Specified)	TurboX C610
	SKU2	2+64GB	No	LA[2)]	C610-D26-369PWXX C610-D26-ACXPW00 (LA Specified)	TurboX C610
	SKU3	4+64GB	Yes	LA	C610-D46-369PWXX C610-D46-ACXPW00 (LA Specified)	TurboX C610
QCS410	SKU4	2+16GB	Yes	LA[2)]	C410-D21-369PWXX C610-D46-ACXPW00 (LA Specified)	TurboX C410
QCS410	SKU5	2+64GB	No	LA[2)]	C410-D26-369PWXX C410-D26-ACXPW00 (LA Specified)	TurboX C410
QCS410	SKU7	4+64GB	No	LA	C410-D46-369PWXX C410-D46-ACXPW00(LA Specified)	TurboX C410

➲ **NOTES:**

[1)] indicates standard SKU series. For standard SKU series, retail stock is available; as to non-standard SKU series, in case of any customer requirement, specific orders need to be placed accordingly.

[2)] indicates that when the memory is less than 4GB, Android 12 OS runs out of memory. In this case, it is necessary to tailor the system as required to stabilize the system.

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Chapter 2. Development Environment Setup

Refer to *TurboX SOM SDK Manager User Guide* to set up the development environment.

➲ **NOTE:** Make sure that you use the sdkmanager built based on Ubuntu 18.04 with the "`bash turbox-sdkmanager-setup.sh --os-version 18.04 -u`" command, as shown in Figure 2-1:

```
$ bash turbox-sdkmanager-setup.sh --os-version 18.04 -u

****************************************************************
Welcome to turbox-sdkmanager-setup.sh version v3.0.0
This script is used to set up Thundercomm TurboX SOM SDK Manager.

Thundercomm TurboX SOM SDK Manager is an all-in-one tool that
bundles developer software and provides an end-to-end development
environment solution for all TurboX SOM SDKs.

This script requires root privileges. Please enter the sudo password.

You can obtain the help information via turbox-sdkmanager-setup.sh -h, or get more details from:
https://docs.thundercomm.com/turbox_doc/documents/common/tools/turbox-sdkmanager-user-guide
****************************************************************


Docker image built based on Ubuntu 18.04 is selected
Pulling docker image...
docker pull  public.ecr.aws/k5o4b3u5/thundercomm/turbox-sdkmanager-18.04:latest
latest: Pulling from k5o4b3u5/thundercomm/turbox-sdkmanager-18.04
Digest: sha256:39eb81bc80666cb2756c5cd2991b851cd1769dbf8b581cdeb795139c53bc546d
Status: Image is up to date for public.ecr.aws/k5o4b3u5/thundercomm/turbox-sdkmanager-18.04:latest
public.ecr.aws/k5o4b3u5/thundercomm/turbox-sdkmanager-18.04:latest
Making a new tag for none docker image...
No need to retag for docker image.
docker_version v3.1.1
The container name turbox-sdkmanager-18.04_v3.1.1_1000 is already in use.
docker start turbox-sdkmanager-18.04_v3.1.1_1000
turbox-sdkmanager-18.04_v3.1.1_1000
docker exec -it turbox-sdkmanager-18.04_v3.1.1_1000 /bin/create_user.sh 1000 1000
turbox:x:1000:1000::/home/turbox:/bin/bash
User turbox(1000) already exists
docker exec -it -u 1000:1000 -w /home/turbox turbox-sdkmanager-18.04_v3.1.1_1000  /bin/bash
[turbox@sdkmanager-18.04-3.1.1 ~ ]$
```

sdkmanager version

OS version that the sdkmanager is built based on

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Figure 2-1.sdkmanager Window

Chapter 3. Get Started with SDK

3.1. SDK information

Table 3-1 presents the LA SDK information, including the SDK version.

Table 3-1. LA SDK information

SDK Version	Baseline	Baseline Distribution	OS Version
LA2.0.Post-CS1.r002002	QCS610.LA.2.0	Post-CS[5] 0.0.013.1	Android 12.00

3.2. SDK block diagram

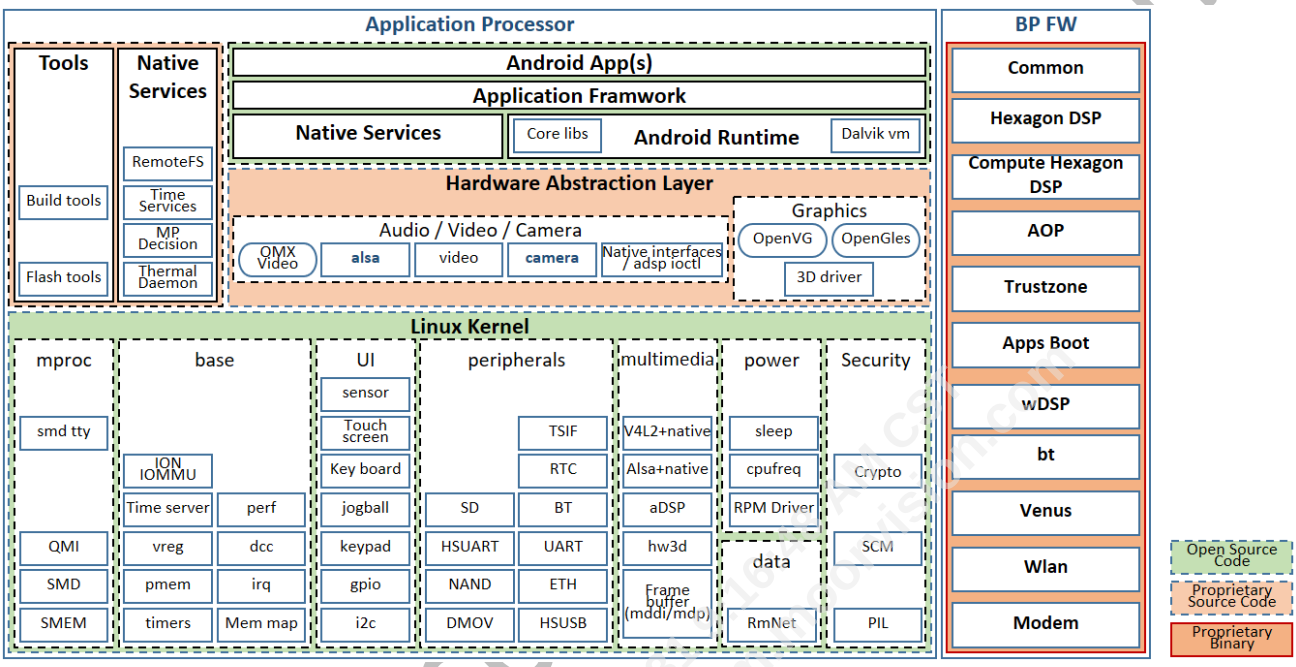


Figure 3-1. Block Diagram

3.3. SDK source tree structure

```
SDK.Turbox_C610.LA.2.0/
├── about.html -> common/about.html          //SDK version info(Product,
Distribution,Components, …)
├── contents.xml -> common/contents.xml      //image contents file
├── common/config/emmc/partition.xml         //eMMC partition file
├── LINUX
│   └── android
│       ├── bionic
│       ├── bootable/bootloader/edk2         //bootloader sources
│       ├── bootable/recovery                //recovery sources
│       ├── build             //basic config files for development kit, such as files for
system build rules and generis
│       ├── dalvik
│       ├── developers              //developer directory
│       ├── development             //infrastructure-related files for Android application
development
│       └── device                       //device related config
│           └── qcom/sm6150              //QCS610 device-related config directory
│       ├── external                     //files related to open-source development kit
│       ├── frameworks                   //source code under Android frameworks
```

```
│       ├── hardware/qcom                //code for QCS610 hardware abstraction layer
 (bootctrl, Display, gps, media, vr, …)
│       ├── kernel
│       │   └── msm-5.4                                  //kernel sources
│       │       ├──arch/arm64/boot/dts/vendor/qcom    //DT source files
│       │       └──arch/arm64/configs/vendor          //kernel config files
│       ├── libcore                                     //core library files
│       ├── libnativehelper          //dynamic library, which is the basis to JNI library
│       ├── Makefile                                 //global Makefile to define build rules
│       ├── packages                                 //application packages
│       ├── pdk                                      //plug development kit
│       ├── platform_testing                         //platform testing files
│       ├── prebuilts                //prebuild resources under x86 and arm framework
│       ├── sdk                                      //sdk and simulator
│       ├── system                   //underlying file system, application, and component
│       ├── test                                     //Android Vendor testing framework
│       ├── toolchain                                //tool chain files
│       ├── tools                                    //tools files
│       └── vendor                                   //QCOM private repository
│           └── qcom
│               ├── opensource
│               └── proprietary
├── turbox_build.sh -> turbox/tools/build_script/common/turbox_build.sh      //SDK build
file.
├── turbox_flash.sh -> turbox/tools/build_script/device/turbox_flash.sh      //Fastboot
flash tool
├── turbox
│   └── tools
│       ├── build_script         //SDK build script(android build, zip, …)
│       └── flash_tools          //Flash tools files(qdl, …)
├── adsp_proc             //aDSP image is released as a binary and no build compilation is
needed
├── aop_proc              //AOP image is released as a binary and no build compilation is
needed
├── boot_images           //BOOT image is released as a binary and no build compilation is
needed
├── btfm_proc             //BTFM image is released as a binary and no build compilation is
needed
├── btfm_proc_CMC         //BTFM_CMC image is released as a binary and no build
compilation is needed
├── cdsp_proc             //cDSP image is released as a binary and no build compilation is
needed
├── modem_proc            //MODEM image is released as a binary and no build compilation
is needed
├── trustzone_images      //TZ image is released as a binary and no build compilation is
needed
├── venus_proc            //VIDEO image is released as a binary and no build compilation
is needed
├── wdsp_proc             //wDSP image is released as a binary and no build compilation is
needed.
└── wlan_proc             //WLAN image is released as a binary and no build compilation is
needed.
```

3.4. Download SDK and image package

Follow the steps below to initialize sdkmanager, download the SDK code and download the image package provided by Thundercomm.

➲ **NOTE:** Please download the SDK code and image package in the sdkmanager.

3.4.1. Initialize sdkmanager

Run `sdkmanager --init` to initialize sdkmanager. Refer to *TurboX SOM SDK Manager User Guide* for details.



Figure 3-1.sdkmanager Initialization Windows

➲ **NOTE:** As shown in Figure 3-1, personal account information is hidden.

3.4.2. Download SDK

Run `sdkmanager --sdk_download` to download the SDK code. Refer to *TurboX SOM SDK Manager User Guide* for more details.



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Figure 3-2.sdkmanager SDK Download Windows

➲ **NOTE:** In the step of "`Product choices are as follows:`", as the serial number varies with the account privilege, please select the correct serial number according to your screen display. If the user specifies the download path, the path directory can only be one level.

3.4.3. Download image package

Run `sdkmanager --image_download` to download the image package, refer to *TurboX SOM SDK Manager User Guide* for more details.



Figure 3-3.sdkmanager Image Download Windows

➲ **NOTE:** In the step of "`Product choices are as follows:`", as the serial number varies with the account privilege, please select the correct serial number according to your screen display.

Customers can also obtain image package provided by Thundercomm from:

- **For debug version:**
  - *FlatBuild_TurboX_C610_xx.xx_LA2.0.R.userdebug.Post-CS1.r002002.zip*
    MD5: 0cd88d14a8bd2598aadd3866b0ad7bc1
  - *FastbootBuild_TurboX_C610_xx.xx_LA2.0.R.userdebug.Post-CS1.r002002.zip*
    MD5: 45b547a8aad16926ea763d6a0bedf4ca

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Chapter 4. Build Images and Pack ROM Package

4.1. Build images

To compile code, run the following script file in the root directory of source code for building debug version.

Please run the commands in the sdkmanager.

```
[turbox@sdkmanager-18.04-3.1.1 ~ ]$ cd workspace/sourcecode/turbox-c610-la2.0-dev.release.Post-CS1.r002002/

[turbox@sdkmanager-18.04-3.1.1 turbox-c610-la2.0-dev.release.Post-CS1.r002002 ]$ ./turbox_build.sh -a -l -v userdebug


[turbox@sdkmanager-18.04-3.1.1 turbox-c610-la2.0-dev.release.Post-CS1.r002002 ]$ ./turbox_build.sh -u -v userdebug
```

➲ **NOTE:** For detailed usage method, input `./turbox_build.sh -h` to view the help information.

4.2. Pack ROM package

Follow the steps below to pack ROM package for debug version.

➲ **NOTE**: Make sure that you run the commands in the sdkmanager.

Step 1. Build image. Refer to 4.1. Build images for more details.

Step 2. Run the following command to pack the ROM package. You can find the ROM package at *turbox/output*.

- Pack **Fastboot** image

  ```
  [turbox@sdkmanager-18.04-3.1.1 turbox-c610-la2.0-dev.release.Post-CS1.r002002 ]$ ./turbox_build.sh --zip_fastboot_build -v userdebug
  ```

- Pack **Flat** image

  ```
  [turbox@sdkmanager-18.04-3.1.1 turbox-c610-la2.0-dev.release.Post-CS1.r002002 ]$ ./turbox_build.sh --zip_flat_build -v userdebug
  ```

➲ **NOTES:**

- Alternatively, customers can obtain ROM package provided by Thundercomm (see also 3.4.3 Download image package).
- For detailed usage method, input `./turbox_build.sh -h` to view the help information.

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Chapter 5. Flash images to board

➲ **NOTES**:

- Stop the ModemManager on your host PC and close the application that occupies the ttyUSB0 port, such as **Minicom**.

```
$ sudo systemctl stop ModemManager
```

- Run the commands in the following sub-sections in the sdkmanager.

5.1. Flash FlatBuild images with 9008

Step 1. Enter the **9008 Mode**.

1) Power off the device by disconnecting the power cable and USB cable.

2) Connect the power cable.

3) Hold the **Force_USB_Boot** button.

4) Press the **Power on** button and wait for about 3 seconds.

5) Release the **Force USB Boot** button and **Power on** button.

6) Connect the Type-C USB cable to an Ubuntu computer.

To check if your device has entered the **9008 Mode**, run the following command:

```
[turbox@sdkmanager-18.04-3.1.1 turbox-c610-la2.0-dev.release.Post-
CS1.r002002 ]$ lsusb
……
Bus 001 Device 042: ID 05c6:9008 Qualcomm, Inc. Gobi Wireless Modem (QDL mode)

……
```

Step 2. Prepare the image package.

- To obtain Thundercomm package, please refer to 3.4.3 Download image package.
- To pack your own image package, please refer to 4.2. Pack ROM Package.

Step 3. Flash the image.

```
[turbox@sdkmanager-18.04-3.1.1 ]$ unzip
FlatBuild_TurboX_C610_xx.xx_LA2.0.R.userdebug.Post-CS1.r002002.zip
[turbox@sdkmanager-18.04-3.1.1 ]$ cd
FlatBuild_TurboX_C610_xx.xx_LA2.0.R.userdebug.Post-CS1.r002002
[turbox@sdkmanager-18.04-3.1.1 FlatBuild_TurboX_C610_xx.xx_LA2.0.R.userdebug.Post-
CS1.r002002 ]$ turbox-flash-flat.sh
```

If the following log appears, it indicates that the flashing progress is successful.

```
16:45:47: INFO:                    (done)
16:45:47: INFO:      | |
16:45:47: INFO:    __| | ___  _ __   ___
16:45:47: INFO:   / _` |/ _ \| '_ \ / _ \
16:45:47: INFO:  | (_| | (_) | | | |  __/
16:45:47: INFO:   \__,_|\___/|_| |_|\___|
```

Step 4. Unplug and plug in the power cable to restart the device.

➲ **NOTE:** For more information of the flash tool, input "`turbox-flash-flat.sh -h`" or access turbox-flash-flat.sh User Guide.

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5.2. Flash images on Linux with Fastboot

Step 1. Enter the **Fastboot** mode.

Option 1. Enter the following commands when the device in **ADB** mode.

```
[turbox@sdkmanager-18.04-3.1.1 ~ ]$ adb root
[turbox@sdkmanager-18.04-3.1.1 ~ ]$ adb reboot bootloader
```

Option 2. Power off the device, hold the **Volume down** button and click the **Power on** button, then release the **Volume down** button about 5 seconds later.

➲ **NOTE**:

To check if the device has entered the **FastBoot** mode, run the following command:

```
[turbox@sdkmanager-18.04-3.1.1~ ]$ fastboot devices
```

The progress is successful with the following output displayed on the shell:

```
21f10525     fastboot
```

Step 2. Flash images.

Execute the commands below to flash images to the board. Replace **<folder name of image package>** with the actual image package name of your product.

```
[turbox@sdkmanager-18.04-3.1.1 ]$ unzip
FastbootBuild_TurboX_C610_xx.xx_LA2.0.R.userdebug.Post-CS1.r002002.zip
[turbox@sdkmanager-18.04-3.1.1 ]$ cd
FastbootBuild_TurboX_C610_xx.xx_LA2.0.R.userdebug.Post-CS1.r002002
[turbox@sdkmanager-18.04-3.1.1
FastbootBuild_TurboX_C610_xx.xx_LA2.0.R.userdebug.Post-
CS1.r002002 ]$ sudo ./turbox_flash.sh --all -r -f -v debug
```

➲ **NOTE**: Enter `./turbox_flash.sh -h` to acquire detailed usage.

If the following log appears, it indicates that the flashing progress is successful.

```
sending sparse 'system' 12/12 (181994 KB)...
OKAY [  5.041s]
writing 'system' 12/12...
OKAY [  0.000s]
finished. total time: 236.409s
Loading complete, Device rebooting ...
rebooting...

finished. total time: 0.051s
```

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Chapter 6. Features

This chapter introduces the software features supported by the DK.

6.1. Presented features

This section describes the key software features that have been included in earlier versions.

- **Kernel**
  - Linux Kernel 5.4
  - 64-bit only
- **OS**
  - Android 12
  - F2FS + FBE
  - Dynamic partition
  - Fastboot support
  - Boot image header version 1
  - Camera 2.0 (HAL 3.4)
  - Keymaster 4.0
  - Vulkan 1.1
  - OTA Android S to S
  - Incremental OTA
- **CV**
  - FastCV
- **Security**
  - Secure boot
  - Secure video encoding
  - Secure camera
  - Secure DSP
- **Connectivity**
  - WLAN 802.11ac
  - USB 3.1 Gen 1
  - Bluetooth 5.0
- **Storage**
  - eMMC
  - Dual rank2 channel LPDDR4x memory support
- **Multimedia**
  - YUV + Bayer capture
  - sHDR v3.8
  - sHDR tuning through Chromatix
  - LDC
  - TNR (MCTF)

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6.2. New features

This section describes the new key software features that are included in this SDK version.

- Upgrade the baseline to QCS610.LA.2.0 Post-CS[5] 0.0.013.1
- Bring up LT6911UXC HDMI-IN
- TF-Lite, OpenGL, SNPE, QNN verification
- UVC/UAC host support
- TurboX software version support
- User build adb root enablement

6.3. On-going features

None.

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Chapter 7. Test Report

➲ **NOTE:** Refer to Table 7-1 for the test results on SKU3 (QC610 4+64GB).

Table 7-1. Sanity test report

Test case	Status
Fastboot flash	Pass
QFIL Flat flash	Pass
QFIL Meta flash	Pass
Recovery mode (adb & volume up)	Pass
Retrieve the abl/xbl/kernel version	Pass
Power on button	Pass
Volume up/down button	Pass
Force USB Boot button	Pass
Fastboot (adb & volume down)	Pass
BT/Wi-Fi/Gen LED	Pass
LSM6DSOTR sensor	Pass
SD Card auto mount/umount/multiple plug & unplug	Pass
Debug UART	Pass
Type A connection	Pass
Interaction between Type A and Type C	Pass
Type C connection	Pass
DP	Pass
LCD touch panel	Pass
MIPI LCD (ili9881c)	Pass
MIPI LCD (ili9881p)	Pass
UAC/UVC host	Pass
DMIC	Pass
Line in	Pass
Audio headphone	Pass
Audio speaker	Pass
BT connection	Pass
Wi-Fi connection	Pass
Battery	NA
POE15W power supply	NA
12V DC power supply	Pass
Charge the battery via type-c	NA
Ethernet_Network port indicator	Pass
DHCP client	Pass
Sleep mode wakeup	Pass

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Test case	Status
Random read/write	Pass
HDMI IN	Pass
IMX577 module capture & video	Pass
IMX415 module capture & video	Fail
QNN	Pass
SNPE	Pass
OpenGL	Pass
FastCV	Pass
TF-Lite	Pass

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Chapter 8. Known Issues

Table 8-1. List of known issues

<table>
<tr><th>Issue No.</th><th colspan="2">Description</th><th>Workaround</th></tr>
<tr><td rowspan="8">SOMC610-176</td><td colspan="2">IMX577 front camera only supports certain resolutions:</td><td rowspan="8">Open the camera app, switch to the rear camera, and set the resolution to "OK". Then switch to the front camera, and you will be able to take photos and record videos using the front camera.</td></tr>
<tr><td>Format</td><td>Test Result</td></tr>
<tr><td>4K</td><td>NG</td></tr>
<tr><td>1080P</td><td>NG</td></tr>
<tr><td>720P</td><td>OK</td></tr>
<tr><td>480P</td><td>OK</td></tr>
<tr><td>VGA</td><td>NG</td></tr>
<tr><td>CIF</td><td>OK</td></tr>
<tr><td>QVGA</td><td>OK</td></tr>
<tr><td>SOMC610-183</td><td colspan="2">IMX415 module camera cannot take photos or record videos.</td><td>NA</td></tr>
<tr><td>SOMC610-177</td><td colspan="2">HDMI-IN audio function cannot work.</td><td>NA</td></tr>
<tr><td>SOM610-165</td><td colspan="2">Connecting HDMI-IN during startup will cause the system to crash.</td><td>Do not connect HDMI-IN during startup; connect it after the system has booted up.</td></tr>
</table>

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Appendix 1. Notices

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